

EXERCISE 7

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- 17.4-1.** Suppose that a queueing system has two servers, an exponential interarrival time distribution with a mean of 2 hours, and an exponential service-time distribution with a mean of 2 hours for each server. Furthermore, a customer has just arrived at 12:00 noon.
- (a) What is the probability that the next arrival will come (i) before 1:00 P.M., (ii) between 1:00 and 2:00 P.M., and (iii) after 2:00 P.M.?
 - (b) Suppose that no additional customers arrive before 1:00 P.M. Now what is the probability that the next arrival will come between 1:00 and 2:00 P.M.?
 - (c) What is the probability that the number of arrivals between 1:00 and 2:00 P.M. will be (i) 0, (ii) 1, and (iii) 2 or more?
 - (d) Suppose that both servers are serving customers at 1:00 P.M. What is the probability that *neither* customer will have service completed (i) before 2:00 P.M., (ii) before 1:10 P.M., and (iii) before 1:01 P.M.?

EX 17.4.3

17.4-3. The time required by a mechanic to repair a machine has an exponential distribution with a mean of 4 hours. However, a special tool would reduce this mean to 2 hours. If the mechanic repairs a machine in less than 2 hours, he is paid \$100; otherwise, he is paid \$80. Determine the mechanic's expected increase in pay per machine repaired if he uses the special tool.

EX 17-6.5

17.4-5. A queueing system has three servers with expected service times of 30 minutes, 20 minutes, and 15 minutes. The service times have an exponential distribution. Each server has been busy with a current customer for 10 minutes. Determine the expected remaining time until the next service completion.

Question 1 (15 points)

For an oral exam, students arrive according to a Poisson distribution with an average of 6 students per hour. There are 3 examiners available for the examination. To each examiner, it takes on average 20 minutes to evaluate a student. If all the examiners are busy, the arriving new student have to wait for their turn. Since the room is small, if there are more than two students waiting, the new arriving students has to come back later. Therefore, there are at most 5 students in the system: 3 taking the exam and 2 waiting.

- b. What is the expected number of students in the room at any point in time?
- a. Suppose it is 9am and the examiners are ready to start the exams. There are no students waiting yet. How long does it take before a student arrives and have to wait to take the exam?